

Bit N Build '26 — UP Regionals

OFFICIAL PROBLEM STATEMENTS & GUIDELINES

Theme: AI-Powered Solutions for Real-World Impact

The following problem statements are engineered for advanced teams, encouraging systems that combine AI reasoning, real-world data/APIs, persistent state, explainability, human-in-the-loop controls, and graceful failure handling.

PROBLEM STATEMENTS

01 **HealthTech: AI-Powered Rural Clinical Navigation & Early Risk Detection**

Rural and semi-urban populations often face fragmented healthcare access. A person may have symptoms, previous prescriptions, lab reports, and limited access to qualified healthcare professionals, but no unified system helps them understand what to do next. Build an AI-powered healthcare navigation system that collects information through text, voice, images/documents, or structured forms and helps users navigate appropriate next steps. The system must not replace doctors or provide definitive diagnoses; it should identify potential risk signals and guide users toward appropriate care.

02 **Social Impact: AI Agent for Welfare Scheme Discovery & Application Assistance**

Government welfare programs can be difficult to discover and understand. Eligibility may depend on age, income, occupation, location, education, family composition, or existing benefits. Build an AI-powered public-benefit navigation system that helps citizens discover potentially relevant schemes and guides them through eligibility checks and application preparation. The system should not merely list schemes; it should explain why a scheme may apply and identify missing information or documents.

03 **AgriTech: AI Farm Decision & Market Intelligence Agent**

Farmers make decisions throughout the crop cycle about sowing, irrigation, fertilizer, pests, harvesting, and selling. These decisions are influenced by weather, soil, crop stage, market conditions, and local context. Build an AI agricultural decision-support agent that combines multiple data sources to provide contextual, explainable recommendations rather than generic farming advice.

04 **Climate & Sustainability: AI Energy Optimization & Demand Response Agent**

Buildings, campuses, factories, and small businesses often have detailed energy data but lack systems that turn it into actionable decisions. Build an AI energy-management agent that analyzes consumption and recommends or executes energy-saving actions while respecting operational constraints.

05 **Education & Employability: AI Skill-Gap & Personalized Learning Agent**

Students often know the role they want but do not know which skills they lack, what projects demonstrate those skills, or how to prioritize learning. Build an AI career agent that converts a student's current profile into a measurable and adaptive roadmap.

06 **FinTech: AI-Powered Personal Finance & Financial Risk Agent**

Individuals and small businesses often have transaction data but struggle to understand spending behavior, cash flow, financial risk, and progress toward financial goals. Build an AI financial intelligence agent that analyzes transaction histories, recurring expenses, income patterns, budgets, and goals to identify anomalies, forecast cash flow, simulate scenarios, and recommend actions. High-impact financial decisions must remain under explicit human control.

07 **Open Innovation: Build Beyond the Boundaries**

Many real-world problems do not fit neatly into a predefined technology domain. Participants are therefore invited to identify a meaningful, clearly defined problem and develop an innovative technology-driven solution across domains like Cybersecurity, Accessibility, Smart Cities, Mobility, LegalTech, GovTech, RetailTech, Media & Information, Disaster Management, Defence & Public Safety, Developer Tools, or Robotics. The focus should be on solving a real problem, rather than simply demonstrating a particular technology.

SUBMISSION GUIDELINES

 **GitHub Repository**
Complete source code with a clean README .md covering architecture, dependencies, setup guide, and execution steps.

 **Presentation (PPT)**
Concise slide deck outlining problem, solution, architecture, tech stack, key innovations, and measurable impact.

 **Demo Video**
Working demonstration of primary features and user flow. Hosted on Google Drive with access set to “Anyone with the link can view.”

Submission Process:
A Google Form will be provided at the end of the hackathon. Teams must submit their team details, project information, GitHub repository link, PPT, and public Google Drive video link through the form.

JUDGING CRITERIA

CRITERIA	EVALUATION FOCUS
Innovation	Originality and creativity of the solution.
Technical Implementation	Quality, functionality, and effective use of technology/AI.
Problem Relevance	Understanding and relevance of the problem being addressed.
Impact & Practicality	Real-world usefulness, feasibility, and scalability.
User Experience	Usability, design, and overall experience.
Presentation & Demo	Clarity of PPT, explanation, and demonstration.

Judges will evaluate the overall quality of the idea, implementation, impact, and demonstration.